

A dynamic mathematical model of an ultra-supercritical coal fired once-through boiler-turbine unit ☆



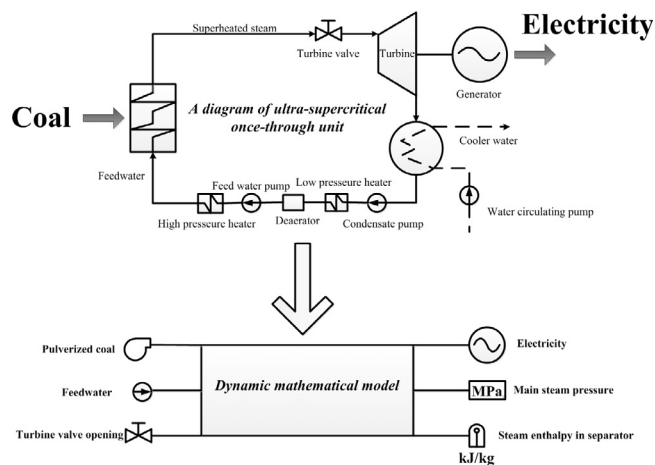
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HIGHLIGHTS

- A mathematical model of an ultra-supercritical unit is developed.
- The analysis of open-loop experiments is described.
- Parameter identification based on immune genetic algorithm is presented.
- The closed-loop validation is performed and compared with the previous study.

GRAPHICAL ABSTRACT



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ABSTRACT

It is challenging and interesting to establish a precise dynamic model of an OTB (once-through boiler) power plant unit in order to meet large scale load demands from the power grid. This study proposes to establish such a dynamic mathematical model of an ultra-supercritical OTB unit under dry operating conditions. More precisely, the dynamic model structure was derived from mass and energy conservation laws as well as thermodynamic principles under some reasonable simplifications and assumptions. Then an IGA (immune genetic algorithm) was improved to identify the parameters, combined with running data. After this, to further enhance model performance, the dynamic mathematical model was extended to be the one with different sets of parameters and functions under different monotonous load ranges. Additionally, open- and closed-loop experiments were conducted in order to further validate the developed model. The experimental results show that the model outputs can approach the actual running data over a wide operating range with appropriate accuracy. More importantly, the dynamic model captures the essential dynamic characteristics of the unit. Therefore, the model can be feasible and applicable for simulation analysis and testing control algorithms.

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Nomenclature

u_B	fuel flow rate command (kg/s)
r'_B	mass flow rate of fuel in the mill (kg/s)
τ	delay time of the pulverizing process (s)
r_B	mass flow rate of fuel in the furnace (kg/s)
M	amount of coal in the mill (kg)
c_0	milling inertia time (s)
c_B	output coefficient of the mill
f_H	correction coefficient of the coal grind ability
f_W	correction coefficient of the moisture content of coal
f_R	correction coefficient of coal fineness
p	steam pressure (MPa)
ρ	density (kg/m ³)
D	mass flow rate (kg/s)
h	specific enthalpy (kJ/kg)
T	temperature (°C)
Q_2	heat transferred from tubes to the medium (kJ/s)
ε	drag coefficient
c	specific heat capacity (kJ/(kg·°C))
m	mass (kg)
k_0	the heat released from burning 1 kg pulverized coal
k_1	the tube constant
k_2	steam turbine gain
n	index coefficient
η	efficiency of the steam turbine
u_t	throttle valve opening ($0 \leq u_t \leq 1$)
N_e	active power (MW)
H_n	heat removed by circulating water (kJ/s)
V_t	total working fluid volume in boiler (m ³)
Q_t	heat absorbed by tube in furnace (kJ/s)

Subscripts

a	average value
in	working fluid in tube
fw	feed water flow
st	throttle steam flow
m	the state in the separator

r	the state in reheater
j	the state in metal
$j1$	the state in metal at outlet of the heated tube
$r1$	the state at inlet of reheater
$r2$	the state at outlet of reheater

Superscript

–	the average value
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Abbreviations

OTB	once-through boiler
DBT	drum boiler-turbine
CCS	coordinated control system
IGA	immune genetic algorithm
GA	genetic algorithm

Units

%	percent
MW	megawatt
g/kWh	gram per kWh
kWh	kilowatt hour
kg/s	kilogram per second
t/h	ton per hour
MPa	megapascal
kg/m ³	kilogram per cubic metre
kJ/kg	kilojoule per kilogram
°C	degree Celsius
kJ/s	kilojoule per second
m ³	cubic metre
s	second
kJ/(kg·°C)	kilojoule per kilogram degree
kg	kilogram
kJ	kilojoule
μm	micron

1. Introduction

Coal-fired power plants play a highly significant role in Chinese power generation, even though renewable energy will be rapidly developed in recent years [1]. Meanwhile, it is estimated that the consumption of China's primary energy will reach 4.8 billion tons of standard coal in 2020, and the fossil fuel can account for 70% of the power demand [2]. In the process of coal burning, many air pollutants may be released, namely, sulfur dioxide (SO₂), nitrogen oxides (NO_x), and inhalable particle whose diameter is less than 10 μm (PM₁₀), extremely dangerous to human health and the global climate [3–7]. Due to the huge environmental pollution caused by coal burning in every year, Chinese government promises to reduce the CO₂ emission per unit of GDP by 60–65% in 2030, compared with that of 2005 [2]. Therefore, it is a pressing need for coal fired power plants in China to find an effective solution to meet above requirements.

Ultra-supercritical coal fired technology is a considerable option, and owns high operating efficiency and low CO₂ emissions per megawatt (MW) generation. According to the statistical data, the operating efficiency of ultra-supercritical units is beyond 45%, more than around 10% compared with the same level of the current subcritical coal fired units in China. Furthermore, in terms of the environmental protection, the release of CO₂ and SO₂ can be reduced by 145 (g/kW h) and 0.4 (g/kW h), respectively [8]. In addition, the ultra-supercritical unit has faster speed of load

response than that of the subcritical unit for less metal content. Nowadays, there are more than eighty ultra-supercritical coal fired plants serving in China, and the plants have prominently significant roles in supplying electricity and maintaining frequency stability of the power network.

Certainly, severe operating conditions have to be in company with a series of technological bottlenecks, such as complex start-up process and less operating reliability. Especially, renewable energy (i.e., wind and solar energy) will be recently energetically developed due to their non-polluting and sustainable merits [2]. Nevertheless, their unstable and random features may pose a threat to the operating stability of the power grid. Therefore, confronted with these pressures, ultra-supercritical coal fired units are obliged to adjust unit load quickly in order to meet the real-time power demand.

In ultra-supercritical units, the CCS (coordinated control system) is used to solve the above problems. However, for any control systems, the satisfactory control performance is dependent on the accuracy and validity of the process model.

In previous studies, the modelling methods can be roughly summarized into three categories. The first is mechanism modelling, which is developed based on fundamental physical laws so as to master the dynamic behaviours of thermodynamic variables [9–16]. Due to the clear physical structure, the state variables in model are observable. Therefore, this kind of model is also called white box model. After obtaining the model structure, it is neces-

sary to optimize the unknown parameters by means of running data and optimization algorithms [17–20]. Particularly, once the mechanism model has been successfully developed, it is able to reflect the rich dynamic behaviours correctly, regardless of various inputs. Due to simplifications and few uncertainties, the accuracy of the white box model is not satisfactory enough. Liu et al. [14] developed an ultra-supercritical once-through unit model with simple structure in mechanism modelling method, in order to meet the requirement for designing the overall control system, through a series of simplifications and assumptions. On the other hand, some studies were also performed on DBT (drum boiler-turbine) units [21–24]. Sunil et al. [25] presented a dynamic model simulator of drum boilers, derived from the first principles and a comprehensive method of combining lumped and distributed modelling used for relevant components.

Then, the second modelling method contains the model using fuzzy [26–28], artificial neural networks [29–32] and neural and fuzzy networks [33–35] strategies. Due to the unclear structure, they are also called the black box model, which can be obtained combined with running data and training procedures. As the operational data is taken fully advantage of, the calculated values can agree well with the measured. However, if confronted with novel inputs, the predicting capability of the black may worsen. Liu et al. [33] presented a model of an ultra-supercritical OTB unit, and two different structures of neural networks (i.e., multilayer perceptron network and neural fuzzy network) were applied to model the unit using running data. The results showed the model built in neural fuzzy network had the satisfactory accuracy and performance.

The third is the grey box model [36–38], a combination of the white and black box model. Considering the few inaccurate relations among variables, some unknown parameters are added in model so as to further capture the unclear relations hidden in running data, and those parameters are finally identified by training procedures like the black. De Ridder and Coomans [37] established a grey-box compartmental model, including the heat loss, internal heat exchange, convection and mixing dynamics. Then, the Markov-Chain-Monte-Carlo method was used for determining the parameters in model.

Due to the clear physical structure, the white box model is firstly chosen in this study. After determining the model structure, the next step is to optimize the unknown parameters.

GA (genetic algorithm) has successfully been used for parameter identification [14,39–42], but it is of the high possibility that the solution estimated by GA converges to the local optima due to the similarity of individuals and weak global searching ability. IGA (immune genetic algorithm) [43,44], a combination of the immune mechanism and genetic algorithm, can effectively enrich the population diversity and avoid local optima. The antibodies, similar to the chromosomes in GA, can promote or restrain themselves effectively to improve population diversity using a selection operator based on the concentration of the antibodies. Moreover, immune selection, immune mutation and the mechanism of the elite reservation can strengthen the ability of global optimization. In short, IGA is more suitable for parameter identification than GA. Xue et al. [45] optimized the enterprise information system structure using an improved IGA based on the binary tree, the niche algorithm and self-adaptive operators. Then the relevant results finally were validated by experiments.

In this research, a dynamic mathematical model was first developed for an ultra-supercritical OTB unit under dry operating conditions. To our best knowledge, there has been no relevant studies except this paper [14] in recent years. Due to its irregular structure and the lack of closed-loop validations, this model can be further improved. Therefore, we established the model with higher accuracy and reasonable structure based on the previous study [14].

The whole outline of the paper is presented as follows. The lumped parameter model of the OTB unit, including the coal-pulverized system and the boiler-turbine system, is developed by means of mass and energy fundamental laws and nonlinear regression analysis in Section 2. During real operation, the parameters in model are varied with time and operating conditions, and many groups of parameters need to be identified. Therefore, the procedure for parameter identification based on the improved IGA is described in Section 3. In Section 4, open-loop experiments by step response are performed to verify the validity of model, and then the calculated outputs are compared against operational data to testify the accuracy. After this, a detailed discussion is presented in Section 5, in order to analyse the reasons behind modelling errors and comparison between this study and the previous [14]. At last, the paper is summarized in Section 6.

2. System description

In this study, a 1000 MW ultra-supercritical power plant unit was researched. The plant adopts single-reheat turbine configuration and tower shape boiler arrangement, introduced from Alstom Power Company. Then there are three superheaters and two reheaters in boiler. Also, there are four low pressure feed water heaters, three high pressure feeder heaters and a deaerator in turbine system, and they can improve cycle thermal efficiency and feed water temperature. Lastly, at nominal condition, the main operating values are shown in Table 1.

2.1. Coal-pulverized system

The coal-pulverized system is a combination of devices and connected tubes, including coal feeder, coal mill, and primary air pipe. The system can transform the raw coal into the pulverized coal, then delivers it to a furnace for suspension combustion. As shown in Fig. 1, the coal feeder can transport the raw coal stored in the bunker to the coal mill through the coal feeder. Then the raw coal can be broken, ground and dried in the coal mill, and the qualified pulverized coal is filtered by the coal powder separator in the coal mill. Lastly, it is directly transported to the furnace for burning.

For the major dynamic behaviours of the coal pulverized system, the main characteristics can be summarized as: time delay and accumulation time.

The time delay mainly existing in the primary air tube and the coal feeder, etc., can be expressed as

$$r'_B = u_B e^{-\tau s} \quad (1)$$

The accumulation time of the coal-pulverized system majorly exists in the coal mill. Thus, the mass balance in mill is

$$dM/dt = r'_B - r_B \quad (2)$$

According to the characteristic of the mill, mass flow rate of pulverized coal can be given by:

Table 1
Plant data at nominal condition.

Item	Value
Nominal power (MW)	1000
Fuel flow (t/h)	408.6
Excessive air ratio	1.2
Feed water temperature (°C)	291.8
Superheated steam flow (t/h)	2924
Steam temperature at superheater outlet (°C)	605
Steam pressure at superheater outlet (MPa)	27.8
Pressure at economizer inlet (MPa)	31.9
Condenser back pressure (MPa)	5.2

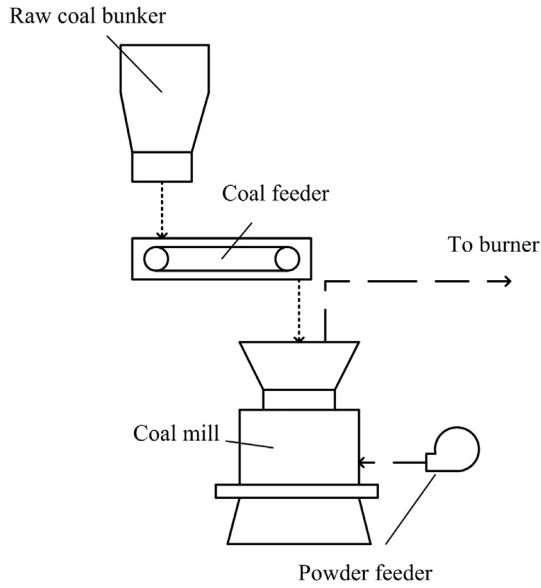


Fig. 1. Simplified diagram of the coal-pulverized system.

$$r_B = c_B M f_H f_W f_R = (1/c_0)M \quad (3)$$

The above product of coefficients - c_B, f_H, f_W, f_R can be substituted for the inverse of c_0 , called milling inertia time. Combined with Eqs. (1)–(3), the transfer function of the coal-pulverized system can be expressed as:

$$r_B = u_B e^{-\tau s} / (c_0 s + 1) \quad (4)$$

2.2. Boiler-turbine system

2.2.1. Simplifications and assumptions

In a boiler-turbine system, the pulverized coal mixed with primary air, burns fully in a furnace to heat the working fluid so as to obtain the steam with high temperature and pressure. Therefore, there are many components in boiler, including economizers, superheaters and reheaters. Afterwards, the qualified steam expands in turbine to work and then generate electricity via generator. The certain simplifications and assumptions are as follows:

1. Flue gas:

- (1) Neglect the axial heat conduction between flue, tube, and working fluid.
- (2) The dynamic change of the flue gas is neglected.
- (3) Heat converted to the tube by gas is proportional to the heat generated by fuel combustion.
- (4) Ignore the variation of the coal quality.

2. Economizer, waterwall, and superheater:

- (1) There are uniform fluid properties in the cross section of tube.
- (2) Regard the economizer, the waterwall, the steam separator and the superheater, as a whole heated tube with the same volume.
- (3) Total feed water flow, including spray water, enters directly into the whole heated tube on the assumption that the superheated steam temperature remains steady within a safe range.

Fig. 2 presents a simplified block diagram of the ultra-supercritical OTB unit, composed of three major sections: the

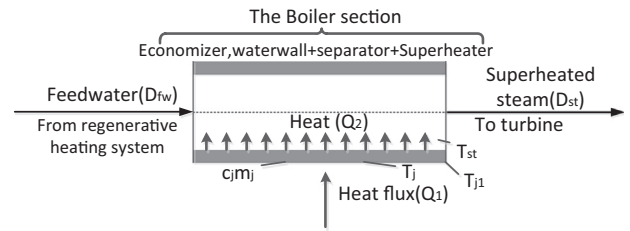


Fig. 2. Simplified diagram of the once-through boiler.

economizer and waterwall, the steam separator, and the superheater. Feed water from a regenerative heating system flows through the heated section, and finally turns into the qualified superheated steam.

2.2.2. The boiler section

To simplify the model structure, this study adopts the lumped parameter method to establish the model. The paper [23] provided that the portion of the energy stored in working fluid is much smaller for a larger plant. Then, the energy stored in metal has great influence on steam during dynamic load variations. The global mass and energy balance equations of the OTB can be written as

$$v_t(d\rho_a/dt) = D_{fw} - D_{st} \quad (5)$$

$$v_t[d(\rho_a h_a)/dt] = D_{fw} h_{fw} - D_{st} h_{st} + Q_2 \quad (6)$$

Remark 1. $h = u + pv$. When 1 kg working fluid flows into a thermodynamic system, thermodynamic energy, u , stored in fluid also enters into the system. The pushing work, pv , received from the external working unit, is also brought into system. Thus, $v_t[d(\rho_a h_a)/dt]$ can represent the change of the energy stored in working fluid.

The metal energy balance equation is

$$\begin{aligned} c_j m_j (dT_j/dt) &= Q_1 - Q_2 \\ c_3 &= c_j m_j \\ Q_1 &= k_0 r_B \\ Q_2 &= k_1 D_{in}^n (T_j - T_{in}) \end{aligned} \quad (7)$$

The function, $Q_2 = k_1 D_{in}^n (T_j - T)$ [50], is the equation of the heat released from metal. The coefficient k_1 , mainly depends on the property of working fluid in condition that the shape of tube inner wall and roughness are constant. In this case, the fluid density is nearly constant, and approximately $n \approx 0.8$.

Remark 2. $c_j m_j (dT_j/dt)$ represents the variation of energy stored in tube metal. Compared with the previous research [14], this study contains the metal energy effect, indicating that the developed model is more reasonable.

It is important for the lumped parameter model to choose a proper lumped parameter point. Generally, the middle point and the outlet point are usually chosen as the representative points. If the middle point is selected, the complexity of model will be enhanced. The other point is much simpler, but it is usually applicable for single-phase fluid due to no phase transition. However, the working fluid in whole heated tube is composed of four phases, namely, unsaturated water, saturated water, saturated steam, and superheated steam. For various states, the difference in thermodynamic property is noticeable. Therefore, it is necessary to select a new position in tube as the lumped parameter point.

The steam pressure in the separator can strongly track load change [21]. Likewise, the steam enthalpy in the separator can reflect the change of the ratio of fuel to water sensitively. Then the ratio needs to be controlled within a safe range to maintain the temperature of superheated steam near the set value. Therefore, the steam state in the separator can be selected as the lumped parameter, and the mass and energy balance equations are:

$$s_1(d\rho_m/dt) = D_{fw} - D_{st} \quad (8)$$

$$s_2[d(\rho_m h_m)/dt] = D_{fw} h_{fw} - D_{st} h_{st} + Q_2 \quad (9)$$

where s_1 and s_2 are dynamic parameters.

Moreover, for convenience, the state at outlet of the tube is chosen as the lumped parameter in metal model. Thus, the metal energy equation is written as

$$c_{j1} m_{j1} (dT_{j1}/dt) = k_0 r_B - k_1 D_{st}^n (T_{j1} - T_{st}) \quad (10)$$

The unit active power can reveal the energy balance between unit plant and power grid, and the main steam pressure is also a signal of balance between boiler and turbine. Additionally, the turbine valve opening can accelerate the speed of the load response, as it can take fully advantage of storage energy to coordinate the dynamic difference between boiler and turbine. The timely control for coal quantity and feed water flow can recover the storage energy so that steam volume can agree well with load demands. Therefore, steam enthalpy and pressure in the separator, fuel flow in a furnace and metal temperature at output of tube are chosen as state variables. Then the steam enthalpy in the separator, the main steam pressure and the unit load are selected as outputs. The fuel quantity order, the feed water flow and the throttle valve opening are regarded as inputs. Eqs. (8) and (9) calculate the partial derivatives with respect to p_m and h_m . Then they can be rewritten as Eqs. (11) and (12)

$$b_{11}(dp_m/dt) + b_{12}(dh_m/dt) = D_{fw} - D_{st} \quad (11)$$

$$b_{21}(dp_m/dt) + b_{22}(dh_m/dt) = D_{fw} h_{fw} - l D_{st} h_m + k_1 D_{st}^n (T_{j1} - T_{st}) \quad (12)$$

where

$$\begin{aligned} b_{11} &= s_1(\partial\rho_m/\partial p_m) \\ b_{12} &= s_1(\partial\rho_m/\partial h_m) \\ b_{21} &= s_2 h_m(\partial\rho_m/\partial p_m) \\ b_{22} &= s_2[h_m(\partial\rho_m/\partial h_m) + \rho_m] \\ l &= h_{st}/h_m \end{aligned}$$

Combined with Eqs. (11) and (12), the following equations can be given by

$$c_1(dp_m/dt) = (h_{fw} - d_1)D_{fw} + (d_1 - lh_m)D_{st} + k_1 D_{st}^n (T_j - T_{st}) \quad (13)$$

$$c_2(dh_m/dt) = (h_{fw} - d_2)D_{fw} + (d_2 - lh_m)D_{st} + k_1 D_{st}^n (T_j - T_{st}) \quad (14)$$

where

$$\begin{aligned} c_1 &= b_{21} - (b_{11}b_{22}/b_{12}) \\ c_2 &= b_{22} - b_{12}b_{21}/b_{11} \\ c_3 &= c_{j1} m_{j1} \\ d_1 &= b_{22}/b_{12} \\ d_2 &= b_{21}/b_{11} \end{aligned}$$

The equation of the differential pressure and the steam flow in superheater section is usually expressed as

$$\Delta p = D_{st}^2 / \varepsilon \quad (15)$$

However, this equation only considers the pressure drop resulting from the mass flow rate, neglecting the factor, the increase of

steam volume flow rate after absorbing heat. Therefore, Tian [46] provided a relatively accurate equation that

$$D_{v0}^2 + z_2 D_{v0} Q_0 = z_1 \Delta p \quad (16)$$

where, D_{v0} is the initial volume flow rate of steam in superheater; $\Delta p = p_m - p_{st}$, is the pressure drop in superheater; z_1 and z_2 are the coefficients relating to tube resistance and specific heat of steam; Q_0 is the heat absorbed by superheated steam flowing through the tube.

According to the equation, $D_{v0} = D_{m0} v_0$, v_0 and D_{m0} are the specific volume and the mass flow rate of steam in the separator, respectively. Eq. (16) can be converted into the following equation

$$Q_0^2 [v_0^2 (D_{m0}/Q_0)^2 + z_2 v_0 D_{m0}/Q_0] = z_1 \Delta p \quad (17)$$

Approximately, D_{m0}/Q_0 can be expressed as a function of the ratio of fuel to water, and h_m can reflect the variation of ratio of fuel to water sensitively. During regular operation, the fluctuation of h_m is relatively small, which can be roughly viewed as a constant. With respect to the energy balance, Q_0 is nearly proportional to the N_e , i.e. the unit active power. Likewise, p_m , i.e. the steam pressure in the separator, can track the change of N_e effectively [21]. Then v_0 can be expressed as a function of p_m and h_m . Therefore, the above equation can be rewritten as

$$\Delta p = g(p_m) \quad (18)$$

where $g(\cdot)$ is a some function.

2.2.3. The turbine section

The equation [47] of the mass flow, density and pressure of the steam at inlet of turbine is

$$D_{st} = \lambda u_i p_{st}^{1-\alpha} \rho_{st}^\alpha, \quad 0 \leq \alpha \leq 0.5 \quad (19)$$

where λ and α are coefficients needed to be determined, and α is related with superheat degree. Its range is from 0 to 0.5, and $\alpha = 0$ corresponds to saturated steam state; also, $\alpha = 0.5$ corresponds to ideal superheated steam state.

As steam density can be expressed as a function of pressure and enthalpy, Eq. (19) can be rewritten as follows:

$$D_{st} = u_i f(p_{st}, h_{st}) \quad (20)$$

where $f(\cdot)$ is a some function of p_{st} and T_{st} .

Fig. 3 shows a simplified block diagram of the turbine and regenerative system.

Compared with boiler section, turbine and regenerative system has a faster dynamic response, and the inertia time is approximately from ten to twenty seconds. The unit power can be expressed as

$$\begin{aligned} N_e &= \eta [D_{st} h_{st} - (q_n h_n + q_{n-1} h_{n-1} + q_{n-2} h_{n-2}) \\ &\quad - D_r h_{r1} + D_{r2} h_{r2} - (q_{n-3} h_{n-3} + \dots + q_0 h_0)] \end{aligned} \quad (21)$$

where q_0 and h_0 are the mass flow rate (kg/s) and specific enthalpy (kJ/kg) of the exhaust steam out of the low pressure cylinder, respectively; q_i and h_i ($1 \leq i \leq n$) are the mass flow rate (kg/s) and

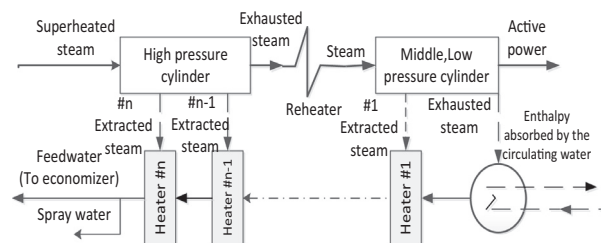


Fig. 3. Simplified block diagram of turbine and regenerative system.

specific enthalpy (kJ/kg) of the extracted steam into heater #*i*, respectively.

Assuming that there is no mass loss in thermal cycle, then Eq. (21) can be rewritten as

$$N_e = \eta[D_{st}h_{st} - (D_{st}h_{fw} + H_n) + D_r(h_{r2} - h_{r1})] \quad (22)$$

In general, H_n is proportional to N_e , i.e., $H_n = gN_e$.

Then

$$N_e = \eta[D_{st}h_{st} - D_{st}h_{fw} + D_r(h_{r2} - h_{r1})] - \eta g N_e \quad (23)$$

Table 2 [48] provides the ratio of heat absorbed by steam in superheater to the one in reheater.

Obviously, with the unit load increasing, the ratio also monotonically increases. Thus, there is a function between unit load and the ratio, which can be expressed as

$$D(N_e) = D_r(h_{r2} - h_{r1})/[D_{st}(h_{st} - h_{fw})] \quad (24)$$

where $D(\cdot)$ is a specific function.

As mentioned earlier, the steam pressure in the separator can track the change of load. Then there is a function between N_e and p_m . Thus, Eq. (24) can be rewritten as

$$P(p_m) = D_r(h_{r2} - h_{r1})/[D_{st}(h_{st} - h_{fw})] \quad (25)$$

During regular operation, h_{fw} , the enthalpy of feed water, also follows the change of the unit load. There is a functional relationship between h_{fw} and p_m , namely, $h_{fw} = h(p_m)$.

Thus, Eq. (23) can be rewritten as

$$N_e = k_2 D_{st}(h_{st} - h_{fw}) \quad (26)$$

where $k_2 = \eta[1 + P(p_m)]/(1 + \eta g)$. That is $k_2 = k(p_m)$, and $k(\cdot)$ is the specific function.

2.3. Summary

Above all, the dynamic nonlinear model can be expressed as

$$\begin{aligned} \dot{X} &= A(X) + B(X)U \\ Y &= C(X) + D(X)U \end{aligned} \quad (27)$$

where

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} r_B \\ p_m \\ h_m \\ T_j \end{bmatrix}, U = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} u_B \\ D_{fw} \\ u_t \end{bmatrix}, Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} p_{st} \\ h_m \\ N_e \end{bmatrix}$$

$$\begin{aligned} \dot{x}_1 &= -(1/c_0)x_1 + (e^{-\tau_s}/c_0)u_1 \\ \dot{x}_2 &= [(h_{fw} - d_1)/c_1]u_2 + [(d_1 - lx_3)/c_1]f(x_2 - g(x_2), lx_3)u_3 \\ &\quad + [k_1(x_4 - T_{st})f^n(x_2 - g(x_2), lx_3)/c_1]u_3^n \\ \dot{x}_3 &= [(h_{fw} - d_2)/c_2]u_2 + [(d_2 - lx_3)/c_2]f(x_2 - g(x_2), lx_3)u_3 \\ &\quad + [k_1(x_4 - T_{st})f^n(x_2 - g(x_2), lx_3)/c_2]u_3^n \\ \dot{x}_4 &= (k_0/c_3)x_1 - [k_1(x_4 - T_{st})f^n(x_2 - g(x_2), lx_3)/c_3]u_3^n \\ y_1 &= x_2 - g(x_2) \\ y_2 &= x_3 \\ y_3 &= k_2(lx_3 - h_{fw})f(x_2 - g(x_2), lx_3)u_3 \end{aligned}$$

Table 2

The ratio of $D_r(h_{r2} - h_{r1})$ to $D_{st}(h_{st} - h_{fw})$ at different steady conditions.

The load (MW)	$D_r(h_{r2} - h_{r1})/[D_{st}(h_{st} - h_{fw})]$
1000	0.198
800	0.195
600	0.191
500	0.183

3. Parameter identification

In this section, there are four unknown static parameters- k_0 , k_2 , l , h_{fw} , seven unknown dynamic parameters- τ , c_0 , c_1 , c_2 , c_3 , d_1 , d_2 , and three unknown functions- $f(\cdot)$, $g(\cdot)$, Q_2 need to identify.

3.1. Static parameter identification

The static parameters can be calculated, combined with nonlinear regression analysis and steady data, which varies within the certain error (i.e. 2% in this paper), namely, staying in steady state.

According to Eqs. (4)–(7), the steady equations can be given by

$$0 = u_B^* - r_B^* \quad (28)$$

$$0 = D_{fw}^* - D_{st}^* \quad (29)$$

$$0 = D_{fw}^* h_{fw}^* - l D_{st}^* h_m^* + k_1 D_{st}^{*n} (T_{j1}^* - T_{st}^*) \quad (30)$$

$$0 = k_0 r_B^* - k_1 D_{st}^{*n} (T_{j1}^* - T_{st}^*) \quad (31)$$

The superscript * represents the variables are in steady state.

Combined with Eq. (26), the calculating equations of static parameters are as follows:

$$l = h_{st}^*/h_m^* \quad (32)$$

$$k_0 = D_{fw}^* (l h_m^* - h_{fw}^*)/u_B^* \quad (33)$$

$$k_2 = N_e^*/[D_{st}^* (h_{st}^* - h_{fw}^*)] \quad (34)$$

$$k_0 u_B^*/(T_{j1}^* - T_{st}^*) = k_1 D_{st}^{*n} \quad (35)$$

k_1 , n can be obtained by nonlinear regression analysis.

Note that the parameters in model, especially dynamic parameters, change with operating conditions and time [49]. Then, the factors, i.e. time and operating conditions, are considered in this study, in order to adapt to practical conditions. Thus, the running data is divided into nine groups to determine static and dynamic parameters and unknown functions according to data characteristic.

The steady-state values are shown in Table 3.

Here, l means the ratio of h_{st} to h_m , namely, determined by heat distribution between waterwall and superheater. Actually, different operating conditions influence the value of l , here, the load section ramps up from 600 MW to 872 MW, and $l = 1.2886$. Moreover, k_0 has corresponding relation with respect to pulverized coal flow in a furnace. By regression analysis, the function between k_0 and the coal quantity in a furnace, r_B , was given by $k_0 = 1.948r_B^2 - 364.56r_B + 32,500$. On the other hand, k_2 , its significance is the unit load that corresponds to 1 kJ energy brought into the turbine by superheated steam. In practice, k_2 follows the change of unit power. According to the steady data, the function of k_2 was determined as $k_2 = 0.0003638p_m^{0.153246}$ by regression analysis.

According to steady data at different conditions, the equation was expressed as

$$Q_2 = 5198.755D_{st}^{0.757}(T_j - \bar{T}_{st}), \bar{T}_{st} = 586.9411$$

Likewise, the function of h_{fw} was given as $h_{fw} = 575.276p_m^{0.24566}$.

According to steady data, the function of Δp was expressed as

$$\Delta p = g(p_m) = 0.04p_m^{1.065} \quad (37)$$

In addition, the equation, referring to the paper [49], was obtained by dynamic running data over a wide operating range, and the 1stOpt, a optimization software. Thus, the function $-f(p_{st}, h_{st})$, was given as

Table 3
Steady-state values at different steady conditions.

N_e (MW)	h_{fw} (kJ/kg)	h_{st} (kJ/kg)	h_m (kJ/kg)	D_{fw} (kg/s)	u_B (kg/s)	D_{st} (kg/s)	P_{st} (MPa)	P_m (MPa)	u_t	T_{st} (°C)	T_j (°C)
601.3	1143.7	3508.4	2746.1	457.7	62.5	453.8	15.7	16.5	0.658	573.8	575.7
781.2	1220.3	3530.9	2755.5	585.9	87.0	586.7	20.1	21.2	0.666	597.6	599.8
855.9	1249.8	3490.6	2722.5	644.2	94.0	643.3	22.5	23.6	0.652	591.0	593.1
896.0	1263.5	3507.7	2775.8	686.4	99.2	688.0	23.4	24.7	0.668	599.9	602.0
920.9	1263.4	3499.2	2677.6	698.9	101.3	698.5	23.6	24.9	0.672	597.5	599.8
950.6	1280.6	3489.0	2708.3	729.8	106.0	731.3	24.7	26.0	0.669	597.5	599.6
1000.9	1299.4	3480.4	2765.4	777.0	112.8	778.9	26.2	27.7	0.668	599.3	601.4

$$f(p_{st}, h_{st}) = 5145 \cdot (-854 + 13.47p_{st} + 0.4h_{st} - 0.003p_{st}h_{st})^{-0.743} \quad (38)$$

3.2. Dynamic parameter identification

In this section, the dynamic parameters were identified, combined with the improved IGA and operational data, and the sampling period is 10 s.

For IGA training procedure, the error E was determined as follows

$$E = \sum_{i=1}^N (|\Delta p_{st}/p_{st0}| + |\Delta p_m/p_{m0}| + |\Delta N_e/N_{e0}| + |\Delta r_B/r_{B0}| + |\Delta h_m/h_{m0}|) \quad (39)$$

Remark 3. Δp_{st} , ΔN_e , Δh_m , Δr_B , Δp_m represent the errors between the calculated and measured values of corresponding state variables. Likewise, p_{st0} , N_{e0} , h_{m0} , r_{B0} , p_{m0} , indicate the variables lying in the nominal condition.

An antibody consists of τ , c_0 , c_1 , c_2 , c_3 , d_1 , d_2 in developed model. Running data over a wide operating range was used for parameter optimization in order to achieve a precise antibody. As shown in Fig. 4, the optimization procedure is composed of three stages in total. The first stage is initializing algorithm configuration. The second stage, including the three operations, i.e. immune selection, immune crossover and immune mutation, similar to GA, called genetic section. The third stage is the most important part during optimization process, called immune section, including high antibody mutation and large scale mutation of optimal antibody. After the mutation, the fitness value and concentration are calculated. Then a certain amount of antibodies are generated from the mutation of the optimal antibody, in order to enhance the diversity of population and global search speed. If the optimal antibody in current generation is equal to the one in former generation, the large scale mutation of the optimal antibody will be executed. The radius of mutation in multi-dimension space gradually increases till escaping the premature point or meeting max iteration times. The operator combines the merits of binary and real number coding. Finally, the new antibodies generate under the inoculation of memory cells and the next evolution continues.

After the closed-loop identification, the dynamic parameters are as follows:

$$\tau = 167, c_0 = 121, c_1 = 1326462, c_2 = 63028, c_3 = 10152, d_1 = 933, d_2 = 2354. \quad (40)$$

The above part is obtained using dynamic running data that ranges from 600 MW to 872 MW, and the complete are listed in Appendix A. The ranges of three inputs are: $u_B \in [50 \text{ kg/s}, 120 \text{ kg/s}]$, $D_{fw} \in [400 \text{ kg/s}, 800 \text{ kg/s}]$, $u_t \in [60\%, 70\%]$.

4. Modelling verification

4.1. Steady state validation

In steady conditions, Table 4 shows the comparison between the measured and the calculated values, and relative errors.

Remark 4. Relative errors refer to the ratio of deviation between the measured and the calculated values to those at the normal condition, i. e. $P_{st0} = 25.36 \text{ MPa}$, $N_{e0} = 1000 \text{ MW}$, $r_{B0} = 120.78 \text{ kg/s}$, $h_{m0} = 2984.32 \text{ kJ/kg}$, and $P_{m0} = 29.86 \text{ MPa}$.

The result, shown in Table 4, illustrated that the general errors were pretty small, and the minimum was 0; the maximum was 3.47%. This indicates that the model has the great steady performance.

4.2. Dynamic model validation

4.2.1. The analysis of open-loop characteristic

Open-loop experiments that step disturbance is acted on every input were performed, in order to validate the developed model, shown in Figs. 5–7.

Remark 5. Considering the fact that, the open-loop experiment is out of the regular closed-loop operation. Therefore, the parameters varied with the operating conditions, $h_{fw} = h(p_m)$ and $k_2 = k(p_m)$, can be viewed as constants staying in corresponding steady conditions.

The analysis of open-loop characteristics is described as follows.

Fig. 5 shows the responses of model under the step disturbance acting on the coal quantity order. After a period of time delay, the heat absorbed by steam gradually increases due to the inertia effect. Then the steam pressure and enthalpy rise up to the corresponding level as well. Therefore, the unit power reaches up to the higher level compared with the original level.

Fig. 6 shows the responses of model under the step disturbance acting on the feed water flow. The steam in heated tube gets squeezed along with step increase of feed water flow. Thus, the more steam is pushed into turbine to work, and the unit load increases. As the pulverized coal flow is unchanged, the load gradually declines to the original level after the squeeze. When the feed water flow increases, the main steam pressure increases subsequently due to squeeze effect. As the volume flow reduces and the main steam temperature drops, the main steam pressure declines to the lower level than the original level. In addition, the length of the unsaturated section increases, and the heat absorbed by steam decreases. Then, the steam enthalpy in the separator declines as well.

Fig. 7 shows the responses of model under the step disturbance acting on the throttle valve opening. Firstly, the main steam pressure declines immediately resulting from abrupt change of the opening. This leads to storage energy quickly releasing so that the decline of the main steam pressure lessens, and the unit power

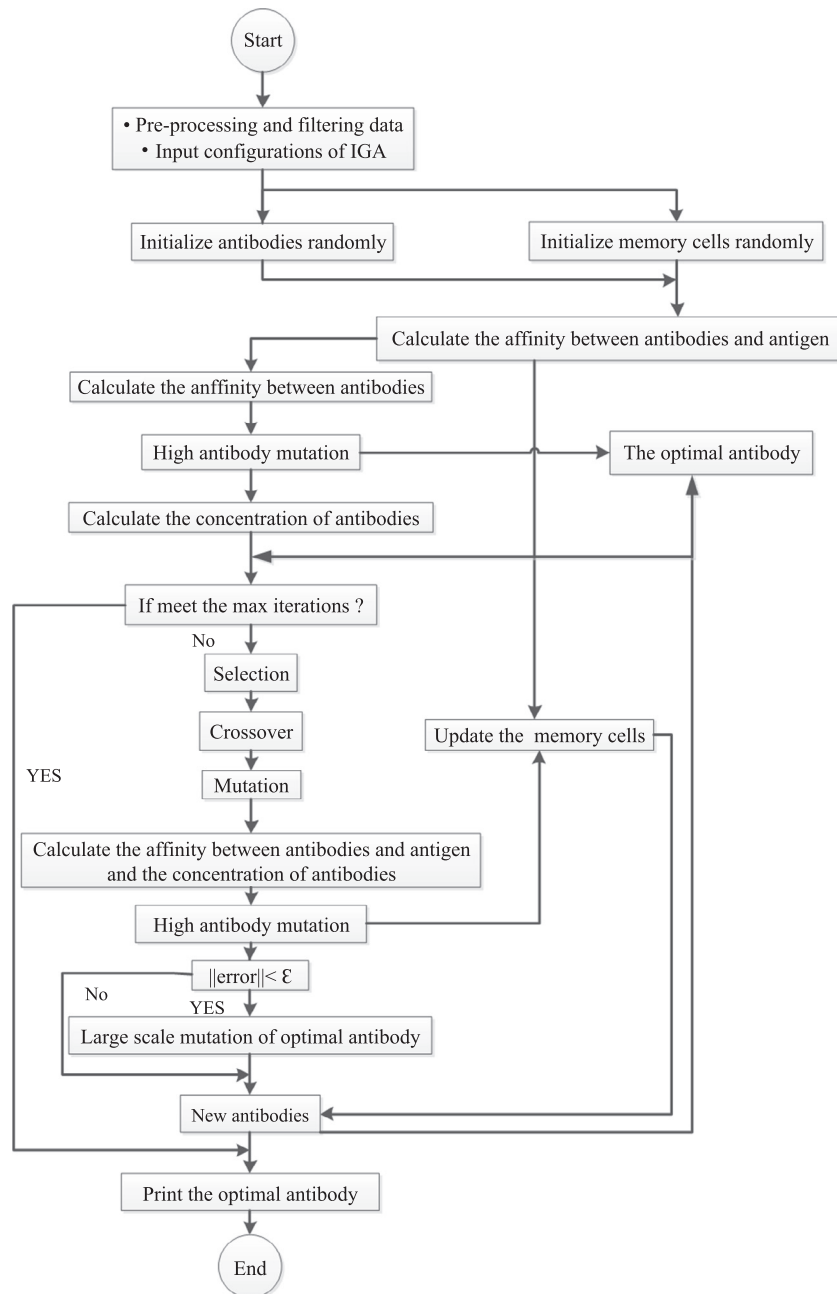


Fig. 4. The training procedure of the OTB model.

increases correspondingly. As the fuel flow rate is unchanged, the unit power recovers to the original level after the release. The main steam pressure gradually decreases to the steady level. Meanwhile, the steam enthalpy in the separator declines slightly owing to the mild variation of steam flow, and lastly recovers to the original level in pace with steam flow due to the unchanged ratio of fuel to water.

From the above analysis, the open-loop dynamic responses of the developed model are reasonable, consistent with the practical operation, indicating the model is correct in essence.

4.2.2. The validation of closed-loop operation

In this section, two sets of the closed-loop running data with different trends are used for model validation. Namely, the load ramps up from 600 MW to 872 MW, and the load ramps down

from 954 MW to 880 MW. Then, the relevant errors are shown in Table 5.

As shown in Figs. 8–13 and Table 5, the calculated values were in strongly accordance with the measured values, and the max relative average error was not beyond 1.27%. This indicates that the developed model owns the satisfactory accuracy under wide dynamic closed-loop operating conditions.

4.2.3. Comparison

In this section, the modelling method was compared with the previous [14], in order to prove the higher accuracy of the developed model. The relevant results are shown in Figs. 14–16, and the average relative errors are listed in Table 6.

As mentioned above, it showed that the calculated values in this research were closer to the measured than those in previous work

Table 4
The comparison between the measured and the calculated values at steady conditions.

Inputs			Measurements						
D_{fw} (kg/s)	u_B (kg/s)	u_t	N_e (MW)	h_m (kJ/kg)	p_{st} (MPa)	p_m (MPa)	r_B (kg/s)		
57.7	62.5	0.658	601.3	2746.1	15.7	16.5	62.5		
585.9	87.0	0.666	781.2	2755.5	20.1	21.2	87.0		
644.2	93.9	0.652	855.9	2722.5	22.5	23.6	93.8		
686.4	99.1	0.668	896.0	2775.8	23.4	24.7	99.2		
698.9	101.3	0.672	920.9	2677.6	23.6	24.9	101.3		
729.8	106.0	0.669	950.6	2708.3	24.7	26.0	106.0		
777.0	112.8	0.668	1000.9	2765.4	26.2	27.7	112.8		
Calculations				Errors (%)					
N_e (MW)	h_m (kJ/kg)	p_{st} (MPa)	p_m (MPa)	r_B (kg/s)	$\Delta N_e/N_{e0}$	$\Delta h_m/h_{m0}$	$\Delta p_{st}/p_{st0}$	$\Delta p_m/p_{m0}$	$\Delta r_B/r_{B0}$
601.9	2717	15.2	15.9	62.5	0.064	0.98	1.98	1.74	0.0260
780.4	2726	19.5	20.5	87.0	0.081	0.99	2.57	2.37	0.0055
852.4	2710	21.8	22.9	93.9	0.347	0.42	2.70	2.26	0.0600
899.3	2786	23.1	24.4	99.2	0.330	0.34	1.16	1.04	0.0000
912.3	2781	23.4	24.6	101.3	0.861	3.47	0.94	0.81	0.0950
945.3	2773	24.6	25.9	106.0	0.529	2.17	0.69	0.56	0.0260
998.4	2768	26.3	27.7	112.8	0.250	0.09	0.09	0.08	0.0110

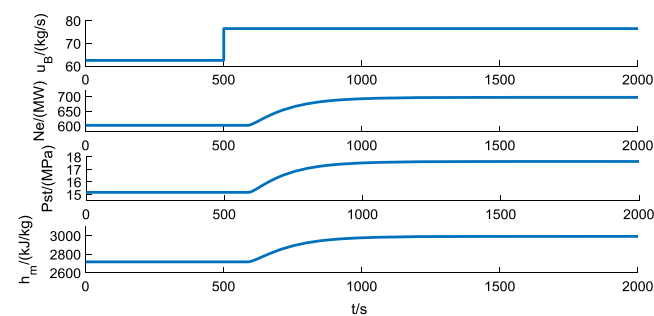


Fig. 5. The step increase of coal quantity order.

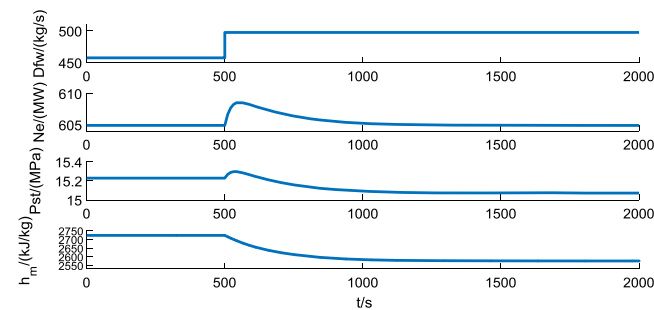


Fig. 6. The step increase of feed water.

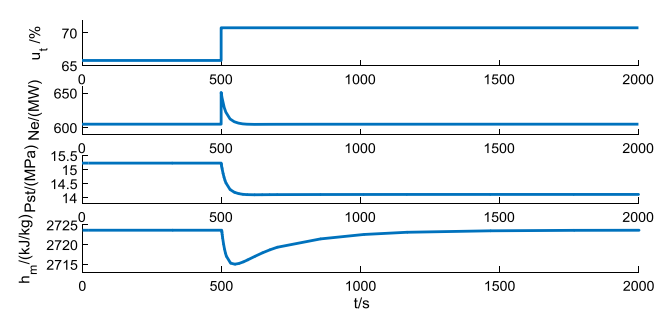


Fig. 7. The step increase of throttle valve opening.

Table 5
Relative error of the calculated outputs.

Range (MW)		$\Delta N_e/N_{e0}$	$\Delta p_{st}/p_{st0}$	$\Delta h_m/h_{m0}$
600–872	Max	2.98	3.31	2.14
	Avg	1.11	0.98	0.81
954–880	Max	4.26	4.64	2.26
	Avg	0.51	1.27	0.91

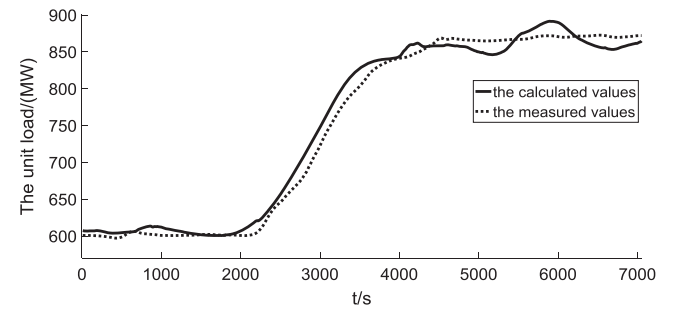


Fig. 8. Comparison on unit load.

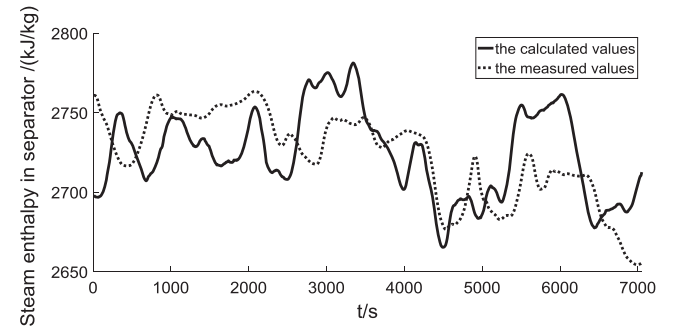


Fig. 9. Comparison on steam enthalpy in separator.

5. Results and discussion

In this section, the model validation under regular operation was investigated and further analysed, and the dynamic operational data from 600 MW to 1000 MW was applied to prove the accuracy and performance.

[14]. The result in table also showed that the average relative errors were reduced by nearly 50%. Therefore, the developed model has higher accuracy.

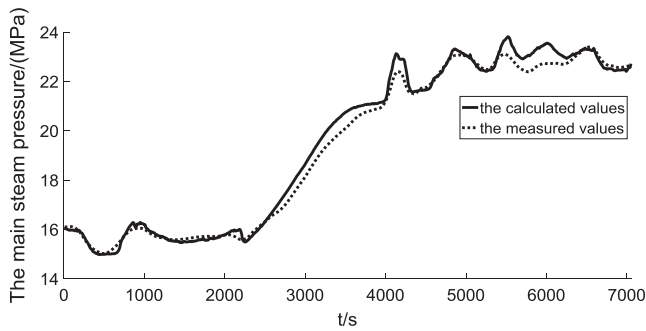


Fig. 10. Comparison on main steam pressure.

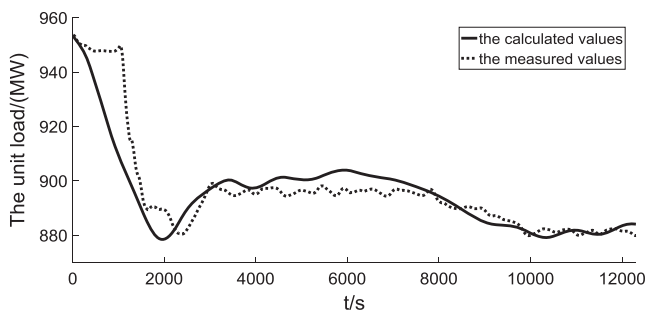


Fig. 11. Comparison on unit load.

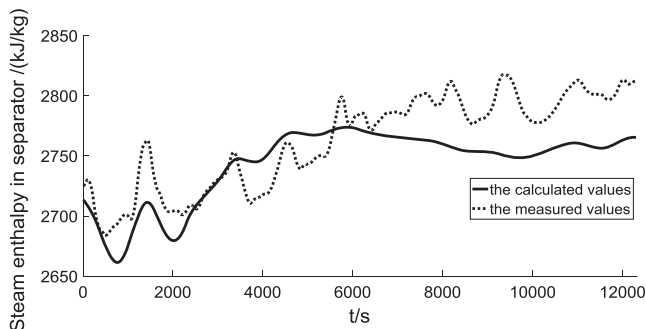


Fig. 12. Comparison on steam enthalpy in separator.

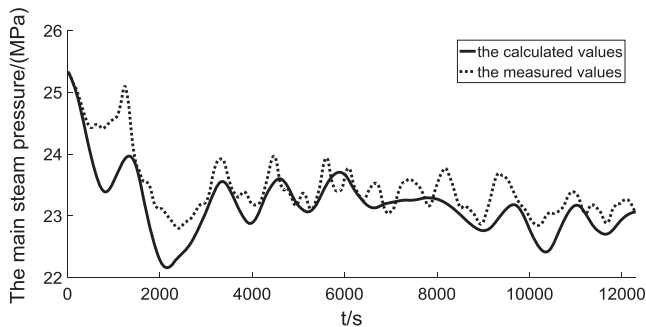


Fig. 13. Comparison on main steam pressure.

As mentioned in Figs. 8–13, the calculated outputs can strongly track the measured values. However, the errors can't be avoided and the reasons are as follows:

First, noise. The noise hidden in the running data needs to be considered. It can unreasonably enlarge the max relative error shown in Table 5.

Table 6

The comparison of two modelling methods in average relative errors.

Method	The average relative error (%)		
	$\Delta P_{st}/P_{st0}$	$\Delta N_e/N_{e0}$	$\Delta h_m/h_{m0}$
The modelling method in this paper	0.93	0.71	0.71
The previous [14]	1.70	1.38	1.65

Second, fitting error. The steady data was applied to determine the functional relations by means of nonlinear regression analysis. In fact, the obtained functions only grasp the major relations with less accuracy. This is why we optimized the nine groups of parameters in order to obtain the higher accuracy.

Third, simplifications and assumptions. For example, dynamic parameters staying in the specific region are viewed as constants and ignore the variation of coal quality, etc.

The modelling method in this paper was compared with the previous [14]. As it is seen in Figs. 14–16 and Table 6, the average relative errors in this study were much less than those in the previous [14] due to the excellent optimization procedure, correct structure and varying parameters, etc. Therefore, the developed model has the higher accuracy and satisfactory performance.

As for applicability, this model is only suitable for the dry operating condition. In this stage, the steam separator is just regarded

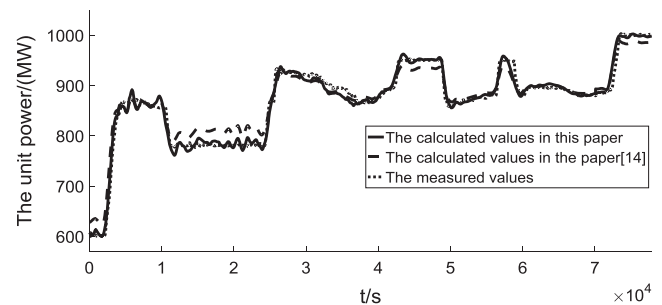


Fig. 14. Comparison on unit load.

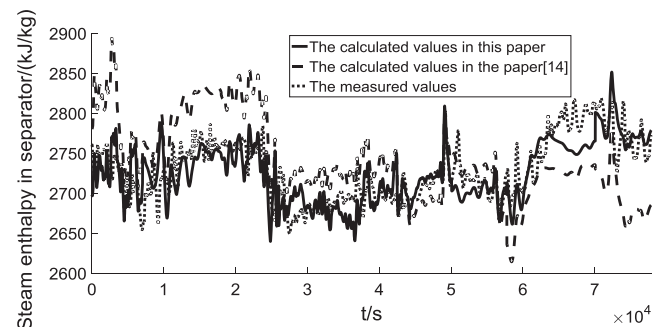


Fig. 15. Comparison on steam enthalpy in separator.

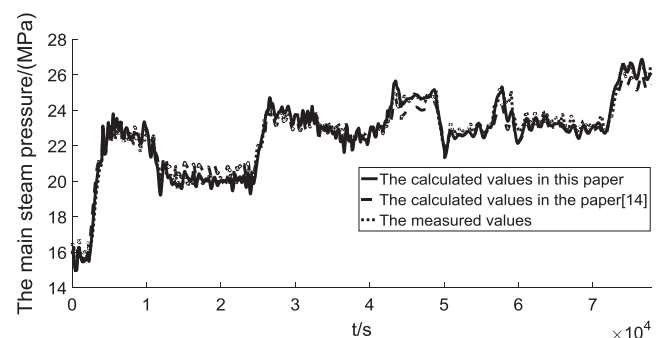


Fig. 16. Comparison on the main steam pressure.

as the tube filled fully with steam. Therefore, the boiler section can be treated as a whole heated tube, and the above modelling analysis is reasonable.

Likewise, in terms of usefulness, the developed model can be used for simulation analysis and testing control algorithms. Due to the reasonable structure and efficient training procedure for parameter identification, the developed model of the OTB unit is able to reflect the dynamic behaviours correctly and accurately under the dry operating condition like open-loop experiments in this study. In addition, due to its correct structure and high accuracy, the model can be viewed as a controlled object to test various control algorithms based on model so as to compare the different control performance and then provide the reference for design of CCS.

Lastly, the major contributions in this study are to put forward a dynamic model of an ultra-supercritical OTB unit and an efficient training procedure for parameter identification. Moreover, the parameters in model are not unique, mainly up to special operational data.

6. Conclusion

In this research, a dynamic mathematical model of the ultra-supercritical OTB unit was developed under the dry operating condition, derived from the mass and energy conservation laws and thermodynamic fundamental principles through reasonable simplifications and assumptions. Afterwards, dynamic parameters were identified, combined with the training procedure based on the improved IGA and running data acquired from an ultra-supercritical plant unit. Through open-loop experiments and closed-loop validations, the developed model can correctly reveal the dynamic characteristics of the OTB unit. After simulation analysis, the structure derived is correct in nature. Then the comparison between the models developed in this study and previous work [14] was performed to prove the higher accuracy under same dynamic operating conditions. Therefore, the model can be used for simulation analysis and testing control algorithms. Additionally, the training procedure for parameter identification was proposed to find an accurate solution so that the model had the satisfactory steady and dynamic performance. Similarly, the procedure can be applied to identify the parameters in other models such as superheaters. In future, the model for an ultra-supercritical OTB unit under wet or start-up conditions, should be investigated.

Acknowledgments

Thanks for the many kind supports and helps from reviewers and editors as well as my roommate, Xiao-ming Wan, regarding paper structure and writing skills.

Appendix A

The complete are as follows.

- (1) The load ramps up from 600 MW to 872 MW, and time range is from 0 s to 7050 s.

- 1) The steady parameters

$$l = 1.2886$$

$$h_{fw} = 575.276p_m^{0.24566}$$

$$k_0 = 1.948r_B^2 - 364.56r_B + 32500$$

$$k_2 = 0.0003638p_m^{0.153246}$$

- 2) Dynamic parameters

$$\tau = 167, c_0 = 121, c_1 = 1326462, c_2 = 63028, c_3 = 10152, d_1 = 933, d_2 = 2354.$$

- 3) Unknown functions

$$\Delta p = g(p_m) = 0.04p_m^{1.065}$$

$$Q_2 = 5198.755D_{st}^{0.757}(T_j - \bar{T}_{st}), \bar{T}_{st} = 586.941$$

$$f(p_{st}, h_{st}) = 5145p_{st}.$$

$$(-854 + 13.47p_{st} + 0.4h_{st} - 0.003p_{st}h_{st})^{-0.743}$$

- 4) Relative errors

$$avg : \Delta p_{st}/p_{st0} = 1.04\%, \Delta N_e/N_{e0} = 1.12\%,$$

$$\Delta h_m/h_{m0} = 0.78\%$$

- (2) The load ramps down from 872 MW to 781 MW, and time range is from 7060 s to 18,000 s.

- 1) The steady parameters

$$l = 1.2847$$

$$h_{fw} = 609.745p_m^{0.2272}$$

$$k_0 = 1.948r_B^2 - 364.56r_B + 32500$$

$$k_2 = 0.0003638p_m^{0.153246}$$

- 2) Dynamic parameters

$$\tau = 228, c_0 = 446, c_1 = 1358150, c_2 = 10713,$$

$$c_3 = 27594, d_1 = 189, d_2 = 1738.$$

- 3) Unknown functions

$$\Delta p = g(p_m) = 0.03865p_m^{1.0755}$$

$$Q_2 = 5198.755D_{st}^{0.757}(T_j - \bar{T}_{st}), \bar{T}_{st} = 595$$

$$f(p_{st}, h_{st}) = 5159.65p_{st}.$$

$$(-854 + 13.47p_{st} + 0.4h_{st} - 0.003p_{st}h_{st})^{-0.743}$$

- 4) Relative errors

$$avg : \Delta p_{st}/p_{st0} = 1.01\%, \Delta N_e/N_{e0} = 0.843\%,$$

$$\Delta h_m/h_{m0} = 0.8237\%$$

- (3) The load ramps up from 781.3 MW to 933.5 MW, and time range is from 18,010 s to 26,010 s.

- 1) The steady parameters

$$l = 1.2845$$

$$h_{fw} = 619.2944p_m^{0.222}$$

$$k_0 = 1.742r_B^2 - 332.8r_B + 31316.7$$

$$k_2 = 0.0004228p_m^{0.10282}$$

- 2) Dynamic parameters

$$\tau = 131, c_0 = 203, c_1 = 1364352, c_2 = 97838,$$

$$c_3 = 35295, d_1 = 1136, d_2 = 1837.$$

- 3) Unknown functions

$$\Delta p = g(p_m) = 0.0329p_m^{1.12764}$$

$$Q_2 = 3917.67D_{st}^{0.797}(T_j - \bar{T}_{st}), \bar{T}_{st} = 596.2$$

$$f(p_{st}, h_{st}) = 5181.22.$$

$$p_{st}(-854 + 13.47p_{st} + 0.4h_{st} - 0.003p_{st}h_{st})^{-0.743}$$

4) Relative errors

$$avg : \Delta p_{st}/p_{st0} = 0.75\%, \Delta N_e/N_{e0} = 0.63\%, \\ \Delta h_m/h_{m0} = 0.69\%$$

(4) The load ramps down from 933.4 MW to 861.5 MW, and time range is from 26,020 s to 37,620 s.

1) The steady parameters

$$l = 1.3 \\ h_{fw} = 573.128p_m^{0.2465} \\ k_0 = 1.195r_B^2 - 246.517r_B + 28044.26 \\ k_2 = 0.0004417p_m^{0.0888}$$

2) Dynamic parameters

$$\tau = 271, c_0 = 276, c_1 = 398652, c_2 = 29544, \\ c_3 = 17080, d_1 = 1191, d_2 = 2151.$$

3) Unknown functions

$$\Delta p = g(p_m) = 0.00837p_m^{1.55317} \\ Q_2 = 4288.73D_{st}^{0.783}(T_j - \bar{T}_{st}), \bar{T}_{st} = 594.3 \\ f(p_{st}, h_{st}) = 5193.23p_{st} \cdot \\ (-854 + 13.47p_{st} + 0.4h_{st} - 0.003p_{st}h_{st})^{-0.743}$$

4) Relative errors

$$avg : \Delta p_{st}/p_{st0} = 0.9\%, \Delta N_e/N_{e0} = 0.91\%, \\ \Delta h_m/h_{m0} = 0.77\%$$

(5) The load ramps up from 861.5 MW to 950.3 MW, and time range is from 37,630 s to 44,240 s.

1) The steady parameters

$$l = 1.2917 \\ h_{fw} = 573.128p_m^{0.2465} \\ k_0 = 1.195r_B^2 - 246.517r_B + 28044.26 \\ k_2 = 0.0004417p_m^{0.0888}$$

2) Dynamic parameters

$$\tau = 290, c_0 = 546, c_1 = 1348217, c_2 = 61678, \\ c_3 = 72647, d_1 = 121, d_2 = 2098.$$

3) Unknown functions

$$\Delta p = g(p_m) = 0.00837p_m^{1.55317} \\ Q_2 = 4288.73D_{st}^{0.783}(T_j - \bar{T}_{st}), \bar{T}_{st} = 591.685 \\ f(p_{st}, h_{st}) = 5194.32p_{st} \cdot \\ (-854 + 13.47p_{st} + 0.4h_{st} - 0.003p_{st}h_{st})^{-0.743}$$

4) Relative errors

$$avg : \Delta p_{st}/p_{st0} = 1.01\%, \Delta N_e/N_{e0} = 0.98\%, \\ \Delta h_m/h_{m0} = 0.96\%$$

(6) The load ramps down from 950.7 MW to 858.4 MW, and time range is from 44,250 s to 50,070 s.

1) The steady parameters

$$l = 1.2846 \\ h_{fw} = 573.128p_m^{0.2465} \\ k_0 = 1.195r_B^2 - 246.517r_B + 28044.26 \\ k_2 = 0.0004417p_m^{0.0888}$$

2) Dynamic parameters

$$\tau = 290, c_0 = 618, c_1 = 828239, c_2 = 35767, \\ c_3 = 44660, d_1 = 106, d_2 = 3381.$$

3) Unknown functions

$$\Delta p = g(p_m) = 0.00837p_m^{1.55317} \\ Q_2 = 4288.73D_{st}^{0.783}(T_j - \bar{T}_{st}), \bar{T}_{st} = 596.3 \\ f(p_{st}, h_{st}) = 5217.4p_{st} \cdot \\ (-854 + 13.47p_{st} + 0.4h_{st} - 0.003p_{st}h_{st})^{-0.743}$$

4) Relative errors

$$avg : \Delta p_{st}/p_{st0} = 0.3\%, \Delta N_e/N_{e0} = 0.34\%, \\ \Delta h_m/h_{m0} = 0.55\%$$

(7) The load ramps up from 858.4 MW to 953.7 MW, and time range is from 50,080 s to 57,760 s.

1) The steady parameters

$$l = 1.2878 \\ h_{fw} = 573.128p_m^{0.2465} \\ k_0 = 1.195r_B^2 - 246.517r_B + 28044.26 \\ k_2 = 0.0004417p_m^{0.0888}$$

2) Dynamic parameters

$$\tau = 78, c_0 = 140, c_1 = 163361, c_2 = 23917, \\ c_3 = 3137, d_1 = 1010, d_2 = 1330.$$

3) Unknown functions

$$\Delta p = g(p_m) = 0.00837p_m^{1.55317} \\ Q_2 = 4288.73D_{st}^{0.783}(T_j - \bar{T}_{st}), \bar{T}_{st} = 594.88 \\ f(p_{st}, h_{st}) = 5198.165p_{st} \cdot \\ (-854 + 13.47p_{st} + 0.4h_{st} - 0.003p_{st}h_{st})^{-0.743}$$

4) Relative errors

$$avg : \Delta p_{st}/p_{st0} = 0.86\%, \Delta N_e/N_{e0} = 0.26\%, \\ \Delta h_m/h_{m0} = 0.58\%$$

(8) The load ramps down from 953.6 MW to 879.9 MW, and time range is from 57,770 s to 70,070 s.

1) The steady parameters

$$l = 1.2638 \\ h_{fw} = 573.128p_m^{0.2465} \\ k_0 = 1.195r_B^2 - 246.517r_B + 28044.26 \\ k_2 = 0.0004417p_m^{0.0888}$$

2) Dynamic parameters

$$\tau = 287, c_0 = 770, c_1 = 905022, c_2 = 18920, \\ c_3 = 20971, d_1 = 716, d_2 = 2512.$$

3) Unknown functions

$$\Delta p = g(p_m) = 0.00837p_m^{1.55317} \\ Q_2 = 4288.73D_{st}^{0.783}(T_j - \bar{T}_{st}), \bar{T}_{st} = 594.4 \\ f(p_{st}, h_{st}) = 5201.16p_{st} \cdot \\ (-854 + 13.47p_{st} + 0.4h_{st} - 0.003p_{st}h_{st})^{-0.743}$$

4) Relative errors

$$\begin{aligned} avg : \Delta p_{st}/p_{st0} &= 1.29\%, \Delta N_e/N_{e0} = 0.56\%, \\ \Delta h_m/h_{m0} &= 0.91\% \end{aligned}$$

- (9) The load ramps up from 879.9 MW to 1001.8 MW, and time range is from 70,080 s to 77,870 s.

1) The steady parameters

$$\begin{aligned} l &= 1.2575 \\ h_{fw} &= 560.92p_m^{0.2531} \\ k_0 &= 1.5134r_B^2 - 358.826r_B + 36239.83 \\ k_2 &= 0.00046p_m^{0.07457} \end{aligned}$$

2) Dynamic parameters

$$\begin{aligned} \tau &= 88, c_0 = 127, c_1 = 819477, c_2 = 10539, \\ c_3 &= 2558, d_1 = 119, d_2 = 1453. \end{aligned}$$

3) Unknown functions

$$\begin{aligned} \Delta p &= g(p_m) = 0.036p_m^{1.1046} \\ Q_2 &= 3833.65D_{st}^{0.8}(T_j - \bar{T}_{st}), \bar{T}_{st} = 598.625 \\ f(p_{st}, h_{st}) &= 5248.91p_{st} \cdot \\ &(-854 + 13.47p_{st} + 0.4h_{st} - 0.003p_{st}h_{st})^{-0.743} \end{aligned}$$

4) Relative errors

$$\begin{aligned} avg : \Delta p_{st}/p_{st0} &= 0.94\%, \Delta N_e/N_{e0} = 0.88\%, \\ \Delta h_m/h_{m0} &= 0.75\% \end{aligned}$$

References

- [1] He Y, Xu Y, Pang Y, et al. A regulatory policy to promote renewable energy consumption in China: review and future evolutionary path. *Renewable Energy* 2016;89:695–705.
- [2] Zhao ZY, Chen YL, Chang RD. How to stimulate renewable energy power generation effectively? – China's incentive approaches and lessons. *Renewable Energy* 2016;92:147–56.
- [3] Zhao Y, Wang SX, Duan L, Lei Y, Cao PF, Hao JM. Primary air pollutant emissions of coal-fired power plants in China: current status and future prediction. *Atmos Environ* 2008;42:8442–52.
- [4] Díaz-Ramírez M, Sebastián F, Royo J, Rezeau A. Influencing factors on NOx emission level during grate conversion of three pelletized energy crops. *Appl Energy* 2014;115:360–73.
- [5] Liu GK, Li ZQ, Chen ZC, Zhu XY, Zhu QY. Effect of the anthracite ratio of blended coals on the combustion and NOx emission characteristics of a retrofitted down-fired 660-MW_e utility boiler. *Appl Energy* 2012;95:196–201.
- [6] Si J, Liu X, Xu M, Sheng L, Zhou Z, Wang C, et al. Effect of kaolin additive on PM_{2.5} reduction during pulverized coal combustion: importance of sodium and its occurrence in coal. *Appl Energy* 2014;114:434–44.
- [7] Loeffler D, Anderson N. Emissions tradeoffs associated with cofiring forest biomass with coal: a case study in Colorado, USA. *Appl Energy* 2014;113:67–77.
- [8] Basu S, Debnath A. Power plant instrumentation and control handbook: a guide to thermal power plants. Academic Press; 2015. p. 875–87.
- [9] Hu Y, Zeng DL, Liu JZ, et al. Dynamic model for controller design of condensate throttling systems. *ISA Trans* 2015;58:622–8.
- [10] Wang D, Zhou Y, Zhou H. A mathematical model suitable for simulation of fast cut back of coal-fired boiler-turbine plant. *Appl Therm Eng* 2016;108:546–54.
- [11] Oko E, Wang M. Dynamic modelling, validation and analysis of coal-fired subcritical power plant. *Fuel* 2014;135:292–300.
- [12] Sun L, Li D, Lee KY, et al. Control-oriented modeling and analysis of direct energy balance in coal-fired boiler-turbine unit. *Control Eng Pract* 2016;55:38–55.
- [13] Shen Y, Yang D, Shen Z, et al. Modeling and analysis of flow instability of the water wall in a 600 MW supercritical W-shaped boiler. *Appl Therm Eng* 2016;99:325–33.
- [14] Liu JZ, Yan S, Zeng DL, et al. A dynamic model used for controller design of a coal fired once-through boiler-turbine unit. *Energy* 2015;93:2069–78.
- [15] Starkloff R, Alobaid F, Karner K, et al. Development and validation of a dynamic simulation model for a large coal-fired power plant. *Appl Therm Eng* 2015;91:496–506.
- [16] Tičá A, Guéguen H, Dumur D, et al. Design of a combined cycle power plant model for optimization. *Appl Energy* 2012;98:256–65.
- [17] Sindareh-Esfahani P, Ghaffari A, Ahmadi P. Thermodynamic modeling based optimization for thermal systems in heat recovery steam generator during cold start-up operation. *Appl Therm Eng* 2014;69(1):286–96.
- [18] Zheng Yu-xin, Liao Ying. Parameter identification of nonlinear dynamic systems using an improved particle swarm optimization. *Optik-Int J Light Electron Opt* 2016;127(19):7865–74.
- [19] Mao Y, Ding F. A novel parameter separation based identification algorithm for Hammerstein systems. *Appl Math Lett* 2016;60:21–7.
- [20] Ferranti F, Rolain Y. A local identification method for linear parameter-varying systems based on interpolation of state-space matrices and least-squares approximation. *Mech Syst Signal Process* 2017;82:478–89.
- [21] Åström KJ, Eklund K. A simplified non-linear model of a drum boiler-turbine unit. *Int J Control* 1972;16(1):145–69.
- [22] De Mello FP. Boiler models for system dynamic performance studies. *IEEE Trans Power Syst* 1991;6(1):66–74.
- [23] Åström KJ, Bell RD. Drum-boiler dynamics. *Automatica* 2000;36(3):363–78.
- [24] Adams J, Clark DR, Louis JR, et al. Mathematical modeling of once-through boiler dynamics. *IEEE Trans Power Apparatus Syst* 1965;84(2):146–56.
- [25] Sunil PU, Barve J, Nataraj PSV. Boiler model and simulation for control design and validation. *IFAC Proc Vol* 2014;47(1):936–40.
- [26] Wu X, Shen J, Li Y, et al. Hierarchical optimization of boiler-turbine unit using fuzzy stable model predictive control. *Control Eng Pract* 2014;30:112–23.
- [27] Wu X, Shen J, Li Y, et al. Data-driven modeling and predictive control for boiler-turbine unit using fuzzy clustering and subspace methods. *ISA Trans* 2014;53(3):699–708.
- [28] Liu X, Kong X. Nonlinear fuzzy model predictive iterative learning control for drum-type boiler-turbine system. *J Process Control* 2013;23(8):1023–40.
- [29] Rostek K, Morytko Ł, Jankowska A. Early detection and prediction of leaks in fluidized-bed boilers using artificial neural networks. *Energy* 2015;89:914–23.
- [30] Dhanuskodi R, Kaliappan R, Suresh S, et al. Artificial neural networks model for predicting wall temperature of supercritical boilers. *Appl Therm Eng* 2015;90:749–53.
- [31] Struśnik D, Golob M, Avsec J. Artificial neural networking model for the prediction of high efficiency boiler steam generation and distribution. *Simul Model Pract Theory* 2015;57:58–70.
- [32] Struśnik D, Golob M, Avsec J. Artificial neural networking model for the prediction of high efficiency boiler steam generation and distribution. *Simul Model Pract Theory* 2015;57:58–70.
- [33] Tunckaya Y, Koklukaya E. Comparative prediction analysis of 600 MW_e coal-fired power plant production rate using statistical and neural-based models. *J Energy Inst* 2015;88(1):11–8.
- [34] Liu XJ, Kong XB, Hou GL, et al. Modeling of a 1000 MW power plant ultra super-critical boiler system using fuzzy-neural network methods. *Energy Convers Manage* 2013;65:518–27.
- [35] Han HG, Ge LM, Qiao JF. An adaptive second order fuzzy neural network for nonlinear system modeling. *Neurocomputing* 2016;214:837–47.
- [36] Jahromi AT, Er MJ, Li X, et al. Sequential fuzzy clustering based dynamic fuzzy neural network for fault diagnosis and prognosis. *Neurocomputing* 2016;196:31–41.
- [37] Afram A, Janabi-Sharifi F. Gray-box modeling and validation of residential HVAC system for control system design. *Appl Energy* 2015;137:134–50.
- [38] De Ridder F, Coomans M. Grey-box model and identification procedure for domestic thermal storage vessels. *Appl Therm Eng* 2014;67(1):147–58.
- [39] Farooq AA, Afram A, Schulz N, et al. Grey-box modeling of a low pressure electric boiler for domestic hot water system. *Appl Therm Eng* 2015;84:257–67.
- [40] Priya K, Babu TS, Balasubramanian K, et al. A novel approach for fuel cell parameter estimation using simple genetic algorithm. *Sustainable Energy Technol Assess* 2015;12:46–52.
- [41] Chisari C, Bedon C, Amadio C. Dynamic and static identification of base-isolated bridges using genetic algorithms. *Eng Struct* 2015;102:80–92.
- [42] Sarode KD, Kumar VR, Kulkarni BD. Embedded multiple shooting methodology in a genetic algorithm framework for parameter estimation and state identification of complex systems. *Chem Eng Sci* 2015;134:605–18.
- [43] Chaibakhsh A, Ghaffari A, Moosavian SAA. A simulated model for a once-through boiler by parameter adjustment based on genetic algorithms. *Simul Model Pract Theory* 2007;15(9):1029–51.
- [44] Chen F, Tang B, Chen R. A novel fault diagnosis model for gearbox based on wavelet support vector machine with immune genetic algorithm. *Measurement* 2013;46(1):220–32.
- [45] Souza SSF, Romero R, Pereira J, et al. Artificial immune algorithm applied to distribution system reconfiguration with variable demand. *Int J Electr Power Energy Syst* 2016;82:561–8.
- [46] Xue C, Dong L, Liu J. Enterprise information system structure optimization based on time property with improved immune genetic algorithm and binary tree. *Comput Math Appl* 2012;63(7):1155–68.
- [47] Liang Tian. Research of unit plant nonlinearity dynamic model. Beijing, China: North China Electric Power University; 2005 (in Chinese).
- [48] Leva A, Maffezzoni C, Benelli G. Validation of drum boiler models through complete dynamic tests. *Control Eng Pract* 1999;7:11–26.
- [49] Yan Shu. A non-linear control model of ultra-supercritical unit. Beijing (China): North China Electric Power University; 2013 (in Chinese).
- [50] Yan Shu, Zeng Deliang, Liu Jizhen. A simplified non-linear model of a once-through boiler-turbine unit and its application. *Proc CSEE* 2012;11:126–34 (in Chinese).
- [51] Chenyue Zhang. Boiler dynamic characteristics and mathematical model. Water Conservancy and Electric Power; 1987. 13 (in Chinese).